Shaunak Manoj Dinkar

AI / ML Engineer

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SUMMARY

Data Science & Analytics postgraduate (Maynooth University) with hands-on experience building and evaluating deep learning and GenAI systems using PyTorch, TensorFlow, and LangChain. Skilled in designing CNN and ResNet-based architectures, retrieval-augmented generation (RAG) pipelines, and rigorous model evaluation methodology, with a strong foundation in Python, SQL, end-to-end ML experimentation, and production software engineering using Git and REST APIs.

EXPERIENCE

React Developer Intern, C-ACE Governance Solutions Pvt. Ltd. 01/2025 - 07/2025 | Pune, India

- Reduced manual content-update effort by ~40% by designing reusable RESTful API integrations and front-end components using React.js, Node.js, Express.js, and MySQL, improving data consistency across systems.
- Automated customer interactions and real-time, location-based service delivery by integrating a Dialogflow chatbot with the Google Maps API, in Git-based, cross-functional workflows.

Software Development Intern, iConcile Technologies Pvt. Ltd. 05/2024 - 06/2024 | Pune, India

- Cut UI rendering time by ~30% by optimising component and state-management logic in a production React front end.
- Automated structured workflow visualisations by implementing the Dagre graph layout algorithm within React Flow, eliminating manual diagram layout work.

EDUCATION

MSc Data Science & Analytics, Maynooth University 2025 - Present | Ireland

MSc Computer Science, Savitribai Phule Pune University 2023 - 2025 | India

CGPA: 9.48 / 10

BSc Computer Science, Savitribai Phule Pune University 2020 - 2023 | India

CGPA: 9.45 / 10

SKILLS

Programming: Python, SQL (PostgreSQL, MySQL), R, JavaScript
Deep Learning: PyTorch, PyTorch Lightning, TensorFlow, Keras, CNNs, ResNet1D
GenAI & NLP: LangChain, Retrieval-Augmented Generation (RAG), LLMs (Llama 3.3), ChromaDB, Hugging Face, Prompt Engineering
Tools: Git, GitHub, Docker

CERTIFICATIONS

- Google Data Analytics Professional Certificate (Google)
- IBM Machine Learning Professional Certificate (Coursera)

PROJECTS

Single-Lead ECG for Structural Heart Disease Detection | *Python, PyTorch, PyTorch Lightning*

- Reached 71% AUROC from single-lead ECG input alone (vs. 80% for 12-lead ECG) by training a ResNet1D architecture with zero-masking and lead-ablation experiments across 12 cardiac conditions.
- Delivered a clinically-rigorous feasibility assessment for wearable ECG devices by building an evaluation pipeline with dynamic threshold optimisation across AUROC, AUPRC, sensitivity, specificity, PPV, and NPV.

Multi-Strategy Retrieval-Augmented Generation (RAG) System | *Python, LangChain, ChromaDB, Groq API*

- Engineered a modular RAG pipeline spanning 6 retrieval strategies (Naive, Reranker, Corrective, Multi-Hop, Graph, Agentic) for PDF-document QA using LangChain LCEL, ChromaDB, and Llama 3.3 70B via Groq API.
- Improved retrieval precision over single-stage similarity search via two-stage retrieval (BAAI/bge-base-en MMR search + BAAI/bge-reranker-base cross-encoder reranking), SHA-256 deduplication, and self-correcting CRAG query rewriting.

CNN-Based Image Classification - CIFAR-10 | *Python, TensorFlow, Keras*

- Reached 80.5% validation accuracy classifying CIFAR-10 images into 10 categories by designing a 3-block CNN (Conv2D-MaxPooling-Dropout) in TensorFlow/Keras with ReLU activations and Adam optimisation.
- Identified batch size 64, 0.001 learning rate, and 3x3 kernels as optimal by systematically ablating depth, kernel size, learning rate, optimiser (Adam vs. SGD), and batch size across 10+ training runs.